IIRSM Training Course Accreditation Submission Brief

Technical Specification — Chainsaw Maintenance & Cross Cutting v1.4, Ground Level

SECTION 1 | Course Specification & Operational Metrics

Parameter	Detail
Official Course Title	Chainsaw Maintenance & Cross Cutting (Ground Level Framework)
Awarding Body Mapping	Independently mapped to UK National Occupational Standards (NOS); submitted for IIRSM accreditation
Publisher & Course Author	Overleaf Publishers Ltd | Author: David J Daniel | First Edition: July 2026
CPD Points Awarded	5 Verifiable CPD Points
Guided Learning Hours (GLH)	16 Hours — technical text, annotated diagrams, and video modules
Directed Online Assessment	1.5 Hours — 45-question randomised multiple-choice examination mapped to NPTC criteria
Independent Self-Study	1.5 Hours — risk evaluation practice, terminology review, and glossary work
Total Qualification Time (TQT)	19 Hours total
Minimum Passing Threshold	80% or higher on the summative examination — unlimited resit attempts permitted
Target Learner Profile	Commercial chainsaw operators, forestry workers, arborists, estate grounds teams, and landscape professionals
Delivery Format	Integrated print + digital: bound manual with companion progressive web app at chainsawcourses.com

SECTION 2 | Digital LMS Quality Gateways & Integrity Controls

To satisfy e-learning accreditation quality standards, the chainsawcourses.com platform implements a layered compliance and verification infrastructure across all seven sequential training modules:

- **Sequential Module Locking:** Each of the 7 modules unlocks only after the preceding module video is fully watched and the associated knowledge quiz is passed at 80% or above. Students cannot skip ahead or access content out of sequence.
- **Anti-Scrubbing Enforcement:** Video timeline scrubbing is disabled for all first-time viewings. Students must engage with the full duration to satisfy GLH requirements and unlock the module quiz. Seek attempt events are logged to the database with timestamp and count.
- **Device-Locked Activation:** Each activation code is single-use and bonds atomically to the first device UUID encountered. A code cannot be used on a second device unless an administrator explicitly resets the bond, preventing credential sharing.
- **Dynamic Identity Watermarking:** A floating watermark showing the student's registered name and email overlays all video content throughout viewing and repositions every 60 seconds, deterring screen-recording and providing clear attribution.
- **Heartbeat Progress Logging:** Video playback positions are written to the database at 30-second intervals, producing a verifiable timestamped engagement trail per student, per module, to support audit and GLH verification.
- **Legally-Binding Digital Waiver:** Before accessing any course content, students complete a liability waiver using a touch or mouse signature canvas. The signed waiver is rendered as a dated PDF with name, email, and signature image, stored persistently and available to administrators.

SECTION 3 | Core Programme Learning Outcomes & Assessment Criteria

The programme spans three teaching units across six Learning Outcomes (LO) with mapped Assessment Criteria (AC), all independently aligned to UK National Occupational Standards for Chainsaw Operations and City & Guilds NPTC assessment parameters. The assessed framework below reflects the content testable in the 45-question summative examination.

UNIT 1 | Occupational Standards, Health & Safety, and Site Risk Evaluation

LO1: Understand the statutory legal framework and personal safety obligations governing chainsaw ground operations.

AC 1.1: Identify the primary duties of care placed on employers and employees under the Health and Safety at Work etc. Act 1974 (HSWA 1974), including the duty to provide safe plant, safe systems of work, adequate information and training, and a safe working environment, and recognise equivalent frameworks applicable to international operators.

AC 1.2: Explain the operational and maintenance obligations mandated by the Provision and Use of Work Equipment Regulations 1998 (PUWER), covering regular inspection schedules, operator competency requirements, suitability of equipment for the task, and mandatory safety feature standards for chainsaws in commercial operations.

AC 1.3: Summarise the substance risk assessment and exposure control requirements under COSHH 2002 as applied to two-stroke petrol-oil fuel blends and alkylate alternatives, bar and chain lubricating oils, lithium-ion battery cell handling and disposal, and inhalation and dermal risks associated with toxic arboreal species including oak processionary moth frass, yew, and certain hardwood dusts.

AC 1.4: Detail the correct CE/UKCA standard class markings required for personal protective equipment: safety helmets with integral visor and ear defenders, cut-resistant chainsaw gloves, chainsaw safety footwear, and Type A versus Type C protective chainsaw trousers, including manufacturer shelf-life limitations and mandatory pre-use inspection and replacement obligations.

LO2: Evaluate environmental site hazards, formulate a risk assessment, and establish robust emergency response protocols.

AC 2.1: Execute a structured 5-step site-specific risk assessment identifying and recording: ground hazards including uneven terrain, root plates, and wet or loose ground; slope and gradient dynamics; proximity of pedestrians, footpaths, and public access points; overhead targets such as power lines, dead branches, and neighbouring trees; and structural vulnerabilities within the planned working area.

AC 2.2: Formulate a complete emergency communication and extraction plan containing: an accurate Ordnance Survey National Grid reference, postal address, What3Words location identifier, site access route and gate codes, designated meeting point for emergency services, and the confirmed location of the chainsaw trauma first-aid kit.

AC 2.3: Identify the mandatory biosecurity cleaning controls required to interrupt transmission of regulated invasive arboreal pathogens between sites, including Hymenoscyphus fraxineus (Ash Dieback), Phytophthora ramorum (Sudden Oak Death), Cryphonectria parasitica (Sweet Chestnut Blight), and Ips typographus (Spruce Bark Beetle), in line with active Forestry Commission and APHA Statutory Plant Health Notices.

UNIT 2 | Power Unit Architecture, Mechanical Integrity, and Component Maintenance

LO3: Analyse the design attributes, operational differences, and integrated safety feature architecture of internal combustion and battery-powered chainsaw platforms.

AC 3.1: Compare the operational risks, charging hazard profile including lithium-ion thermal runaway triggered by impact deformation, moisture ingress, or incorrect charger use, battery state management requirements, and commercial performance limitations of battery-powered units against internal combustion petrol platforms across a range of forestry and arboricultural contexts.

AC 3.2: Map and explain the mechanical function of the 10 core integrated safety components spanning the front, centre, and rear architecture of a standard chainsaw: Front Hand Guard and Inertia Chain Brake, Exhaust Muffler with Spark Arrestor, Low-Kickback Bar Nose Loop Profile, Chain Catcher, Anti-Vibration Isolation Mounts, On/Off Kill Switch, Throttle Lockout Interlock Trigger, Mandatory Safety Decals, Right-Hand Rear Guard Flare, and Bar Scabbard.

LO4: Diagnose, service, and maintain the full mechanical integrity of a commercial chainsaw cutting assembly, including air system, fuel system, drive components, and cutting chain.

AC 4.1: Detail the correct daily air filtration housing cleaning procedure and interpret spark plug electrode colour indicators to assess in-cylinder combustion health: a light-tan or coffee-brown electrode indicates a correct mixture; grey or white indicates a lean condition or overheating; sooty black indicates a rich mixture, oil fouling, or air filter blockage requiring immediate rectification.

AC 4.2: Explain safe carburettor performance adjustment protocols, identifying the function and factory limit constraints of the Idle speed (LA/T), Low-speed (L), and High-speed (H) needle screws, and explain why unskilled adjustment of the H screw risks catastrophic lean seizure at operating temperature, and why over-enriching risks uncontrolled RPM runaway conditions.

AC 4.3: Differentiate between Rim and Spur (Star) drive sprockets, identify correct wear replacement thresholds at 0.5mm groove depth, and diagnose guidebar rail wear conditions including lateral burr formations requiring dressing, rail channel splaying reducing chain retention, and thermal bluing indicating chronic overheating from inadequate chain oil flow.

AC 4.4: Identify chain pitch, gauge, and drive-link count specifications using the chain identification chart; distinguish between Full-Chisel and Semi-Chisel tooth geometries and their respective cutting performance, durability, and dulling characteristics in clean versus dirty or frozen timber; and calculate correct sharpening profiles applying top-plate filing angles in the 25-to-35-degree range, side-plate depth gauge clearances using a raker tool, and an 85-degree wave hook parameter on the side plate.

UNIT 3 | System Startups, Operational Proving, and Ground-Level Timber Processing

LO5: Implement safe pre-start inspection protocols and machine startup methodologies in accordance with manufacturer guidance and industry best practice.

AC 5.1: Execute a structured pre-start visual inspection covering: chain tension correctness, chain sharpness and tooth integrity, chain brake engagement and release, guidebar straightness and groove cleanliness, chain oiler function with reservoir level, correct fuel mixture or full battery charge, air filter condition and seating, handguard and chain catcher integrity, and exhaust muffler security. Differentiate between the safe cold-start floor-anchor technique (saw on ground, foot through rear handle) and the upright knee-clamp warm-start method.

AC 5.2: Perform the mandatory 4-point pre-use dynamic check immediately before commencing work: confirm chain brake engages and chain stops when bar guard is pushed forward and releases when pulled back; verify oil spray dispersion to bar and chain using a spray-line surface test; confirm chain does not creep at engine tick-over idle; confirm the on/off kill switch cuts the engine immediately when operated.

LO6: Apply mechanical principles to safely identify, manage, and resolve tension and compression forces during ground-level timber cross-cutting operations.

AC 6.1: Analyse a downed log or supported stem to determine where compression (closing/crushing) and tension (opening/splitting) forces reside within the wood fibre structure, and select the correct cutting entry face and sequence to eliminate guidebar pinching, uncontrolled timber movement, and splitting hazards.

AC 6.2: Explain the physical difference between a pulling chain cut on the bottom bar face and a pushing chain cut on the top bar face, and apply this understanding to prevent guidebar jams, reactive machine movement, and loss of operator control.

AC 6.3: Define the upper nose quadrant kickback danger zone of the guidebar; explain the rotational inertia mechanism that triggers reactive kickback when the bar nose contacts wood or is pinched under load; and describe how to execute a controlled plunge bore entry cut safely by engaging first with the lower bar face and pivoting through while avoiding nose contact with the kerf walls.

AC 6.4: Sequence and execute safe cross-cutting workflows for: standard ground-supported logs; oversized timber requiring multiple-pass reduction; advanced step cuts for supported log ends; box splits for timber under extreme compression; snedding branch removal applying correct foot positioning and lower-bar-face technique; and extreme tension scenarios using the Toast Rack graduated reduction or sink-cut method to release stored energy incrementally before final severance.

AC 6.5: Identify the severe hazard vectors when processing windblown windfalls, including stored energy in bent or partially uprooted root plates, unpredictable stem roll direction on slope, entangled canopy tension release, and risk of rapid ground movement. Establish a clear pre-planned escape route at 45 degrees to the rear before committing any cuts to the root plate connection or main stem.

SECTION 4 | Integrated Field Tools — Supplementary App Features (Non-Assessed)

In addition to the seven sequential training modules, the chainsawcourses.com platform provides the following standalone professional tools, accessible to all activated students. These are practical operational aids and reference resources that are entirely non-assessed.

- **Pre-Start & Pre-Use Inspection Checklist:** 12-item pre-start check and 4-item pre-use dynamic check, logged to the database with saw identifier and timestamp. Failed items are flagged; records are reviewable by administrators. Students retain a personal inspection history with the ability to amend entries and add notes.
- **Dynamic Risk Assessment Tool:** GPS-derived site location auto-populating an address, OS National Grid reference, and What3Words identifier. Operators record task and site descriptions, access routes, meeting points, and first-aid kit location, then complete an editable 11-item hazard matrix — kickback, entanglement, manual handling, trips and falls, trapped bars, rolling timber, HAVS, fuel fire, weather, bystander exclusion zones, and lone working — with Likelihood x Severity risk ratings generating Low, Medium, or High scores.
- **Biosecurity & Hazard Zone Map:** Interactive Leaflet map using OpenStreetMap tiles with toggleable illustrative hazard overlays in two categories: Direct Operator Health Hazards (Oak Processionary Moth, Brown-tail Moth) and Statutory Containment Zones (Spruce Bark Beetle, Ash Dieback, Sudden Oak Death, Sweet Chestnut Blight, Western Hemlock Dieback). Selecting a zone displays impact information and mandatory biosecurity control measures. Students are directed to verify active boundaries with live Forestry Commission and APHA notices before operations.
- **Chain Identification Chart:** Searchable quick-lookup tool reproducing the manual appendix at page 137. Select brand — Oregon, Stihl, or Husqvarna — and enter a drive-link number to retrieve chain pitch, gauge, correct file diameter, and top-plate filing angle, plus the full reference table, pitch-to-engine-power guide, and chain designation letter code glossary.
- **Timber Species Reference Guide:** Comprehensive reference covering key UK native and non-native tree species encountered in forestry and arboricultural operations. Each species entry details: cutting difficulty and grain characteristics, firewood quality grade, required seasoning duration, spit risk rating, commercial timber value and applications, traditional crafts uses, active pest and disease threats (including Ash Dieback and Oak Processionary Moth), and operator safety warnings for toxic species and hazardous wood dusts. Supports informed operational planning and on-site decision-making.
- **Cross-Cutting Scenario Simulator:** Interactive SVG-based tension and compression decision trainer presenting three realistic log loading scenarios: sag (downward bow under self-weight), hog (upward bow or crown loading), and sidebend (lateral lean). Students select the intended cut position on the log diagram; a correct entry produces audio success feedback, while an incorrect choice triggers a pinch-error warning. Reinforces LO6 AC 6.1 decision-making in a low-risk digital environment before field application.
- **Industry News Feed:** Curated forestry and arboriculture industry news sourced from approved RSS feeds and managed through the admin dashboard. Administrators approve, reject, or add articles before publication. Students can opt in to push notification subscriptions, receiving breaking news alerts directly to their device even when the app is not open — keeping practitioners current with legislation changes, biosecurity alerts, and equipment recalls.

SECTION 5 | Scope Limitation & Theory-Only Boundary Notice

- **Theory-Only CPD Provision:** This programme provides theoretical knowledge verification and structured CPD hours only. Completion of the digital course and summative examination does not certify a learner as a qualified chainsaw operator for commercial or professional use.
- **No Practical Competency Certification:** The mandatory statutory requirement for face-to-face practical training and independent assessment delivered by a registered professional — such as City & Guilds NPTC CS30 or CS31 — remains entirely separate from and unaffected by completion of this qualification.

SECTION 6 | Advanced Workshop Enrichment Content (Non-Assessed)

The following content is included in the manual and companion app for professional enrichment and advanced field awareness only. This material carries the Advanced Workshop Extension Module designation in the printed text and is entirely distinct from the assessed LO/AC framework — it will not appear in the summative examination.

1. **Two-Stroke Combustion Cycle & Fuel Calculation:** The Suck-Squeeze-Bang-Blow combustion sequence and standard 50:1 fuel-to-two-stroke-oil volume mixing calculations are provided for background mechanical understanding only and are not part of the assessed syllabus.
2. **Oil Pump & Clutch Assembly Overhaul:** Extraction of the centrifugal clutch shoe assembly to expose the underlying automated bar oil pump worm drive, pickup tube, and metering adjuster for advanced maintenance operations.
3. **Advanced Windblown Tree Processing:** Managing complex energy vectors in tangled root-plate windfalls, multi-directional stem loading analysis, and safe sequenced canopy clearance operations beyond ground-level cutting scope.
4. **Winch Rigging & Multi-Redirect Pulley Systems:** Configuring heavy-capacity winch systems with multi-redirect pulley blocks and anchor rigging to stabilise large root-plates and control stem fall direction before committing cuts.

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